

Isolation and Characterization of Ocular Pathogenic Bacteria Associated with Eyeglasses

Mrittika Rani Mondol, Md. Robiul Hasan, Mim Islam, Md. Sadaab Hassan, Md. Abid

Hassan, Nowshin Anjum Diba, Fatematuz Johra, Khondokar Nasirujjaman*

Molecular Health Science Laboratory, Department of Genetic Engineering and Biotechnology, University of Rajshahi, Rajshahi 6205, Bangladesh

*Corresponding author's email: khondokar@gmail.com

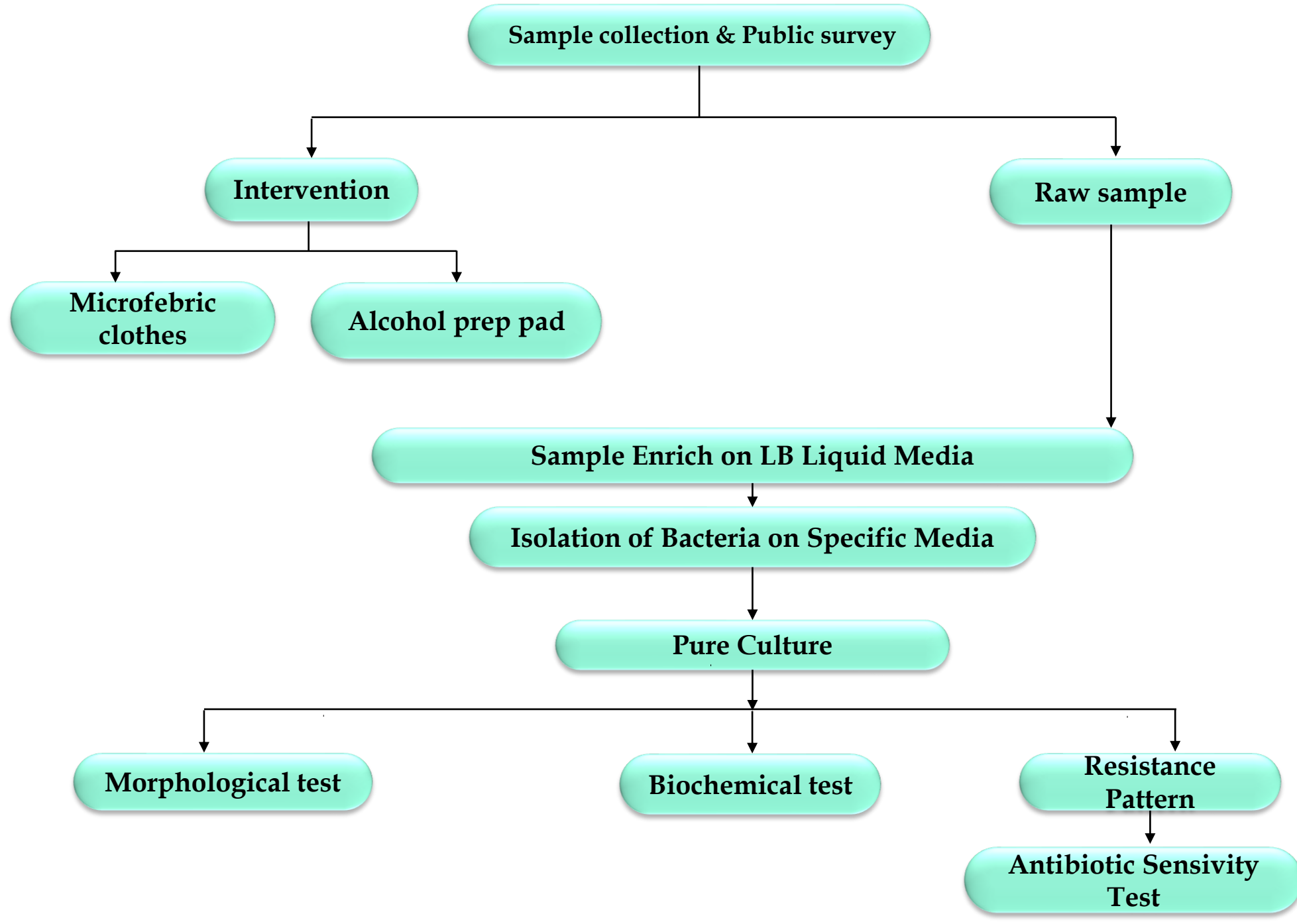
Abstract

The human eye is a major sensory organ and possesses a unique anatomical structure that renders it susceptible to many infectious agents, such as bacteria, including *Staphylococcus epidermidis*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*, and many more. These bacteria contaminate eyeglasses and are transmitted into the eyes resulting in conjunctivitis, keratitis, corneal ulcers, endophthalmitis, and dacryocystitis. The goals of this study were to study the eyeglass cleaning behaviors of users and the prevalence of ocular pathogens in eyeglasses. In this study, bacterial samples were collected from the eyeglasses and a survey was conducted among diverse users. In total, eleven bacterial isolates were selected using Mannitol Salt Agar and Eosin Methyl Blue Agar media and characterized using different biochemical and morphological tests including Gram staining, Motility test, Triple sugar iron test, antibiotic sensitivity tests, etc. The survey findings revealed that significant number of individuals were unfamiliar about presence of germs in glasses and do not properly clean their glasses which can lead to the transmission of ocular pathogenic bacteria resulting in the occurrence of bacterial infections in the eyes. An investigation was conducted on the efficacy of alcohol-based wipes and microfabric-based cleaning procedures where alcohol-based wipes showed best prevention against pathogenic cleaning. In antibiogram study, the majority of the isolated bacteria showed susceptibility only to vancomycin and kanamycin, whereas all isolates showed resistance only to azithromycin. The findings of this study suggest that eyeglasses are potential carriers for germs that might cause eye infections. So, it is necessary to keep them clean using appropriate cleaning materials.

Introduction

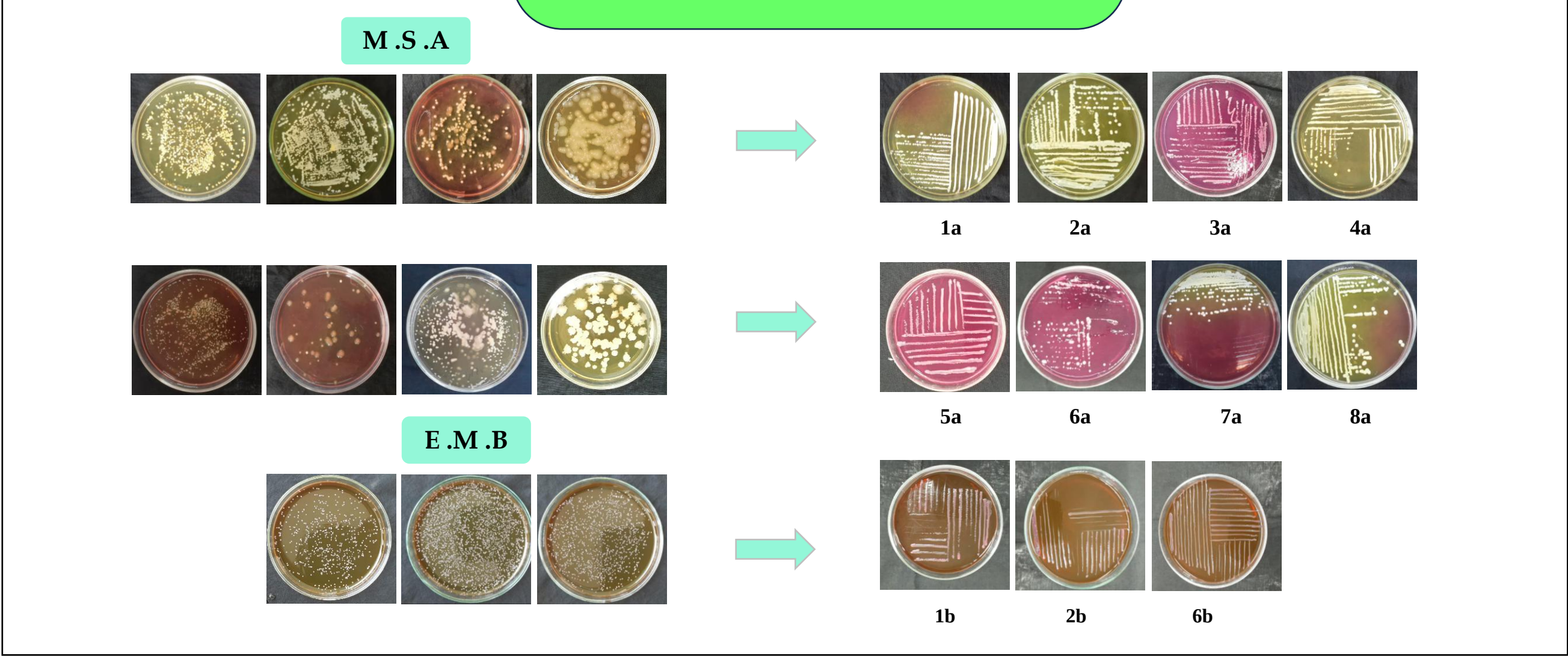
The eye is a major sensory organ that acts somewhat like a camera. Due to its constant exposure to the outdoors it is equipped with many layers of protection such as the eyelid, conjunctiva, and cornea are often susceptible to bacterial invasion (1). *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, and *Proteus mirabilis*, etc. are common ocular pathogenic bacteria that cause conjunctivitis, keratitis, and corneal ulcers (2). More than 4 billion individuals worldwide use glasses to improve their vision and prevent debris and particles from entering their eyes. But these eyeglasses frequently come into contact with our skin and the external environment. They act as fomites, harbor bacteria, and are very risky for eye health. Previous studies have identified pathogenic bacteria such as *E. coli*, *S. aureus*, and *K. pneumoniae* from the surface of eyeglass (3). But no such investigation has been performed in our country. The purpose of this study is to find bacterial contamination in the eyeglasses used by local people and to raise awareness.

Materials and Methods



Results and Discussion

Isolation of Bacteria



Biochemical and Morphological Test Results



Results of Intervention



Antibiotic Sensitivity Test

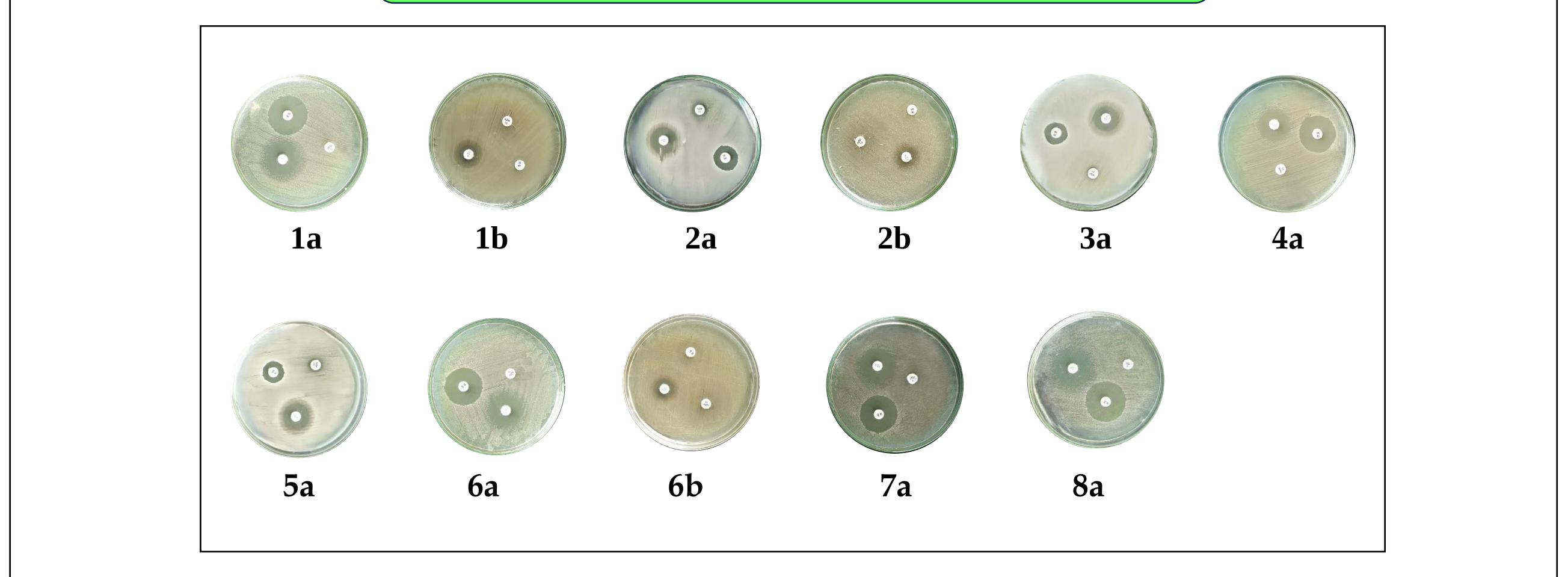


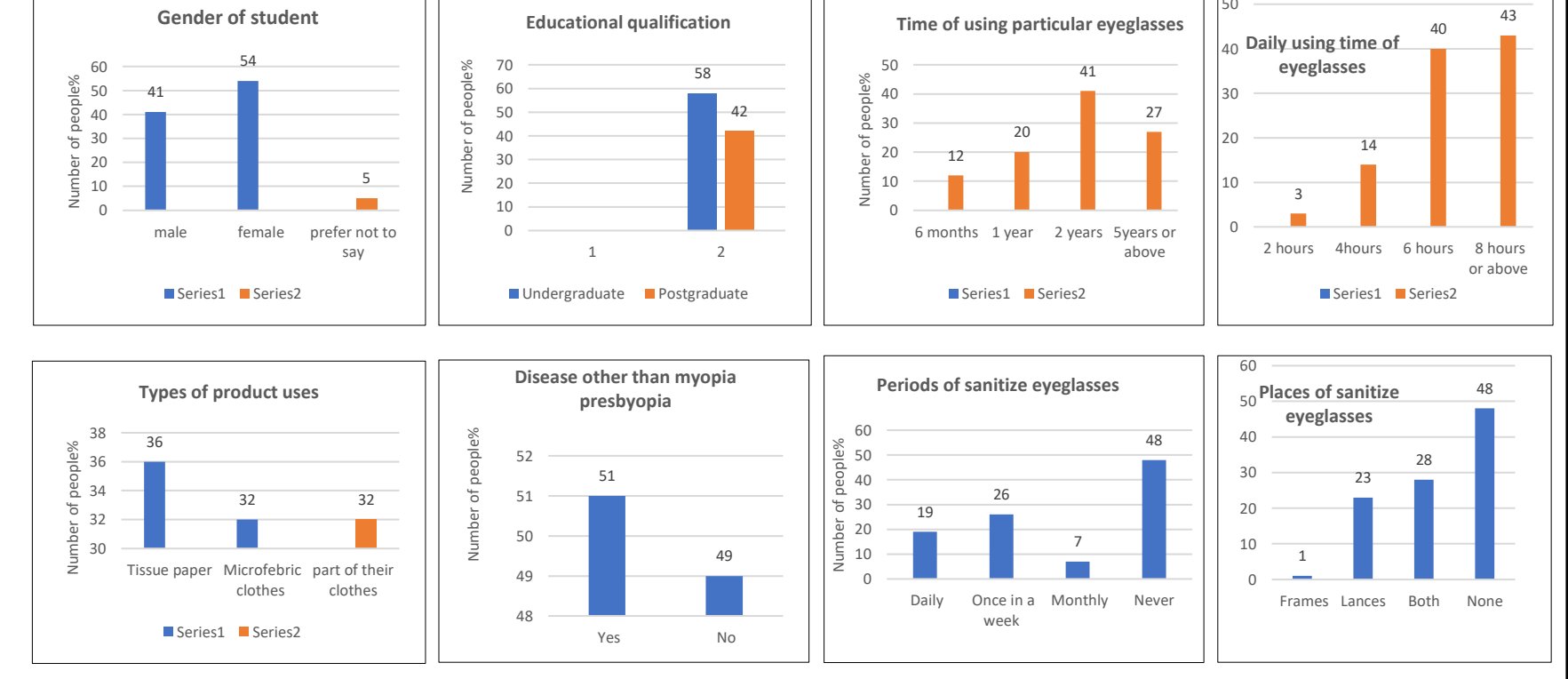
Table-1. Results of Antibiotic Sensitivity Test

Name of Antibiotics (µg/disc)	Zone of Inhibition(mm) and Sensitivity Pattern											
	Isolate Names											
	1a	1b	2a	2b	3a	4a	5a	6a	6b	7a	8a	
Kanamycin (K)-30	16	10	17	12	22	22	20	16	12	20	18	
Vancomycin (VA)-30	15	6	14	6	26	24	18	23	6	24	15	
Azithromycin (AZM)-15	6	6	6	6	6	6	6	7	6	7	6	

Table-2. Summary of Biochemical and Morphological Test

Test	Iso 1a	Iso 1b	Iso 2a	Iso 2b	Iso 3a	Iso 4a	Iso 5a	Iso 6a	Iso 6b	Iso 7a	Iso 8a
Morphological Test											
Gram Staining	+	-	+	-	+	+	+	+	-	+	+
Motility Test	-	-	-	-	-	-	-	-	+	-	-
Biochemical Test											
MacConkey	-	+	-	+	-	-	-	-	+	-	-
Starch Hydrolysis Test	-	-	-	-	-	-	-	-	-	-	-
Simmons Citrate Test	+	+	+	+	-	+	-	-	+	+	+
TSI Test	+	+	+	+	+	+	+	+	+	+	+
Catalase Test	+	-	+	-	+	+	+	+	-	+	+
Indole Test	-	-	-	-	-	-	-	-	-	-	-
Methyl Red Test	+	-	+	-	+	+	+	+	-	+	+
Oxidase test	-	-	-	-	-	-	-	-	-	-	-

Survey Results



Summary and Conclusion

Eyeglasses have the potential to behave as a storehouse for microorganisms that are responsible for recurrent eye infections. The purpose of this study was to explore the communities of bacteria that are present on eyewear that have been worn. After successfully isolating eleven bacterial isolates from worn glasses, morphological and biochemical tests were carried out on the samples. Each and every eyeglasses that were examined was shown to be infected with germs. There was a significant amount of *Staphylococcus* spp that dominated the bacterial population. The presence of *Klebsiella* spp in used eyewear is another concern.

References

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